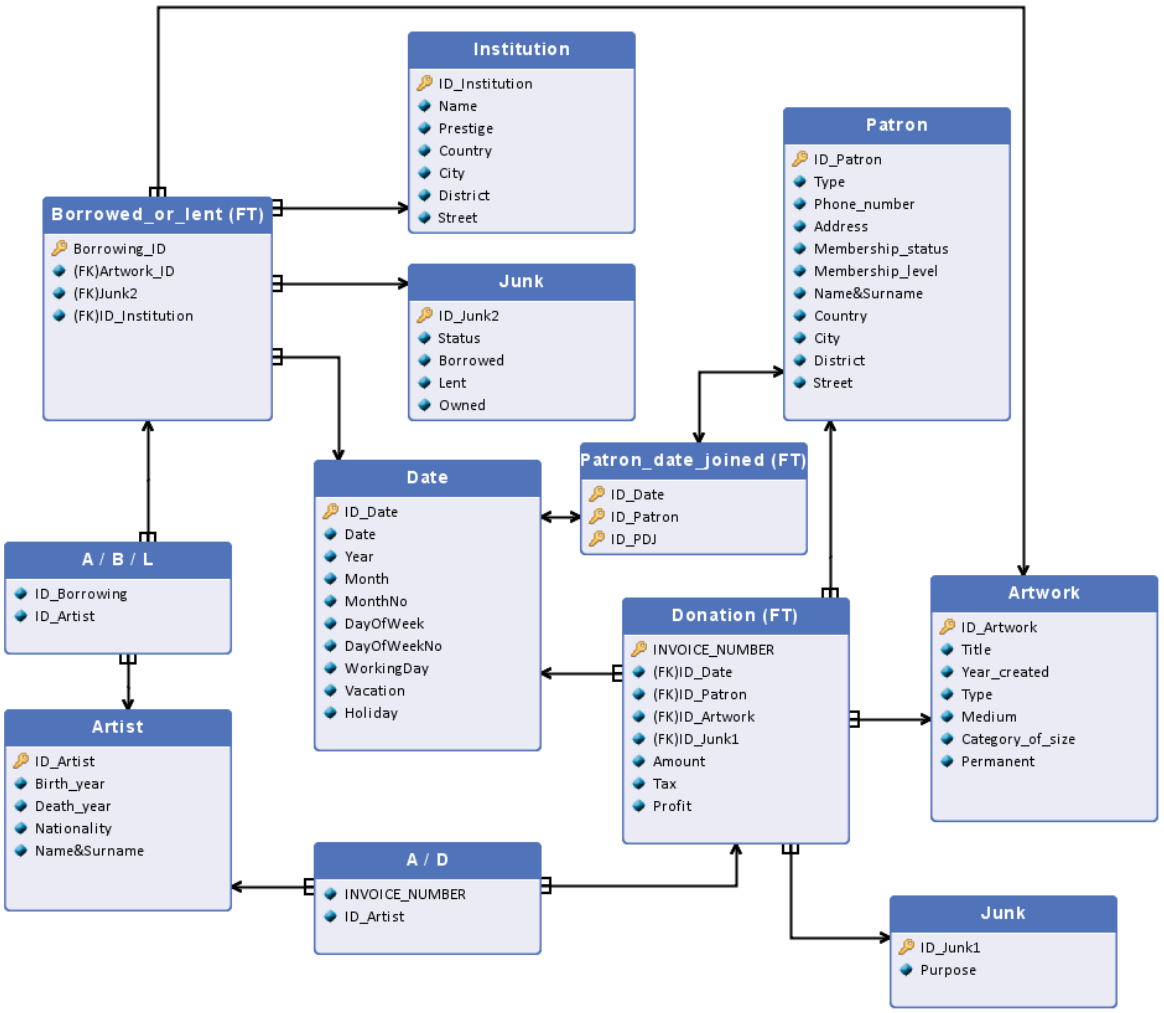
Tates Museum - warehouse design

Business process

Designed for donation business process described in ‘Specification of business process in Tate’s Museum



Donation (Fact Table)		
ATTRIBUTE	**TYPE**	**DESCRIPTION**
INVOICE_NUMBER	Varchar(30)	Unique donation identifier.
ID_Date	Intiger	Date of donation.
ID_Patron	Intiger	Donor key.
ID_Artwork	Intiger	Artwork the donation supports.
ID_Junk1	Intiger	Donation purpose junk dimension.
Amount	Decimal	Amount donated.
Tax	Decimal	Tax portion of the donation.
Profit	Decimal	Profit qual to amount after tax.

Borrowed_or_lent (Fact Table)		
ATTRIBUTE	**TYPE**	**DESCRIPTION**
Borrowing_ID	Intiger PK	Unique transaction ID.
ID_Artwork	Intiger FK	Artwork ID.
ID_Junk2	Intiger FK	ID to junk container.
ID_Institution	Intiger FK	Institution involved in borrowing/lending.

Patron_date_joined (Fact Table)		
ATTRIBUTE	**TYPE**	**DESCRIPTION**
ID_Date	iger Composite k	FK to date, part of composite key
ID_Patron	iger Composite k	FK to patron, part of composite key
ID_PDJ	iger Composite k	ID of patron_date_joined, part of composite key

Artwork (Dimension Table)		
ATTRIBUTE	**TYPE**	**DESCRIPTION**
ID_Artwork	Intiger	Unique artwork identifier.
Title	Varchar(200)	Artwork title.
Year_created	Year	Year the artwork was created.
Type	Varchar(50)	Type (Painting, Sculpture, etc).
Medium	Varchar(500)	Material used.
Category_of_size	Varchar(15)	Based on dimensions (big, medium, small)
Permanent	Boolean	TRUE if part of the permanent collection.

Artist (Dimension Table)		
ATTRIBUTE	**TYPE**	**DESCRIPTION**
ID_Artist	Intiger	Unique artist ID.
Birth_year	Year	Year of birth.
Death_year	Year/NULL	Year of death (NULL if alive).
Nationality	Varchar(100)	Artist's nationality.
Name&Surname	VARCHAR(150)	Full name and surname of the Artist

Patron (Dimension Table)		
ATTRIBUTE	**TYPE**	**DESCRIPTION**
ID_Patron	Intiger	Unique donor ID.
Type	Varchar(50)	"Individual" or "Organization".
Phone_number	Varchar(15)	Contact number.
Address	Varchar(255)	Mailing address.
Membership_status	Varchar(50)	Active, inactive, suspended.
Membership_level	Varchar(50)	Bronze, Silver, Gold, Platinum.
Date_joined	Date	When the donor joined.
Name&Surname	VARCHAR(150)	Full name and surname of the patron
Country	VARCHAR(50)	Country of patron's residance
City	VARCHAR(50)	City of patron's residance
District	VARCHAR(50)	District of patron's residance
Street	VARCHAR(50)	Street of patron's residance

Institution (Dimension Table)		
ATTRIBUTE	**TYPE**	**DESCRIPTION**
Name	Varchar(100)	Institution name.
Location	Varchar(150)	Location of the institution.
Prestige	Numeric	Prestige score (1-10, 10 is the highest).

Date (Dimension Table)		
ATTRIBUTE	**TYPE**	**DESCRIPTION**
Date	Date	Date.
Year	Numeric(4)	Calendar year.
Month	Varchar(10)	Full month name.
MonthNo	Numeric	Month number (1–12).
DayOfWeek	Varchar(10)	Weekday name.
DayOfWeekNo	Numeric	Weekday number (1 = Monday).
WorkingDay	Bool	True if working day.
Vacation	Varchar(20)	e.g., non-holiday, summer holiday.
Holiday	Bool	True if during holiday.

Junk_Donation (Dimension Table)		
ATTRIBUTE	**TYPE**	**DESCRIPTION**
ID_Junk1	Intiger	Surrogate junk key.
Purpose	Varchar(255)	Purpose of donation (restoration, acquisition, etc.)

Junk_Borrowing (Dimension Table)		
ATTRIBUTE	TYPE	DESCRIPTION
ID_Junk2	Intiger	Surrogate junk key.
Status	Varchar(50)	e.g., Ongoing, Completed.
Borrowed	BOOLEAN	Indicates if the artwork is borrowed (TRUE/FALSE)
Lent	BOOLEAN	Indicates if the artwork is lent out (TRUE/FALSE)
Owned	BOOLEAN	Indicates if the artwork is owned (TRUE/FALSE)

A / D (Container to avoid many to many)		
ATTRIBUTE	TYPE	DESCRIPTION
INVOICE_NUMBER	Intiger	Donation ID (FK).
ID_Artist	Intiger	Artist ID (FK).

A / B / L (Container to avoid many to many)		
ATTRIBUTE	TYPE	DESCRIPTION
ID_Borrowing	Intiger	Borrowing ID (FK)
ID_Artist	Intiger	Artist ID (FK).

Fact Definitions

Fact 1: Donation Fact

Description:
A donation of a specified amount by a specified patron for a specific artwork, made on a given day, with a defined purpose, linked to the artwork's ownership and exhibition status.

Fact Table: Donation

Granularity:

- A specific donation (INVOICE_NUMBER)
- A specific date of donation

- A specific patron
- A specific artwork
- A specific purpose of the donation
- Artwork ownership/exhibition status (borrowed/lent/owned)

Measures and Aggregation Functions:

- Number of donation facts – COUNT(1)
- Total amount donated – SUM(Amount)
- Tax – SUM(Tax)
- Profit – SUM(Profit)
- Number of distinct donors – DISTINCT COUNT(ID_Patron)

Fact 2: Borrowing/Lending Fact

Description:
A transaction in which an artwork is borrowed, lent, or confirmed as owned by the museum

Fact Table: Borrowed_or_Lent

Granularity:

- A specific borrowing/lending transaction
- A specific artwork

- A specific institution
- A specific loan status (ongoing/completed, borrowed/lent/owned)

Measures and Aggregation Functions:

- Number of borrowing/lending facts – COUNT(1)
- Number of artworks lent – SUM(Lent)
- Number of artworks owned – SUM(Owned)

Dimension Definitions

Dimensions for Fact 1: Donation

DIMENSION / ATTRIBUTE	TABLE.COLUMN	TYPE
Donation Number	Donation.INVOICE_NUMBER	Degenerate dimension
Artwork	Artwork	Dimension

Title	Artwork.Title	Dimension attribute
Year created	Artwork.Year_created	Dimension attribute
Type	Artwork.Type	Dimension attribute
Medium	Artwork.Medium	Dimension attribute
Category_of_size	Artwork.Category_of_size	Dimension attribute
Permanent	Artwork.Permanent	Dimension attribute
Artwork ID	Artwork.ID_Artwork	Surrogate key

Patron	Patron	Dimension
Type	Patron.Type	Dimension attribute
Phone number	Patron.Phone Number	Dimension attribute

Address	Patron.Address	Dimension attribute
Membership status	Patron.Membership_status	Dimension attribute
Membership level	Patron.Membership_level	Dimension attribute
Date joined	Patron.Date_joined	Dimension attribute
Name	Patron.Name&Surname	Dimension attribute
Country	Patron.Country	Dimension attribute
City	Patron.City	Dimension attribute
District	Patron.District	Dimension attribute
Street	Patron.Street	Dimension attribute
Patron ID	Patron.ID_Patron	Surrogate key

Date of Donation	Date	Dimension
Year	Date.Year	Dimension attribute
Month	Date.Month	Dimension attribute
MonthNo	Date.MonthNo	Dimension attribute
Day of Week	Date.DayOfWeek	Dimension attribute
Working Day	Date.Working Day	Dimension attribute
Holiday	Date.Holiday	Dimension attribute
Date ID	Date.ID_Date	Surrogate key

Donation Purpose	Junk	Junk Dimension
Purpose	Junk.Purpose	Dimension attribute
Junk1 ID	Donation.ID_Junk1	Surrogate key
Artist	Artist	Dimension
ID_Artist	Artist.ID_Artist	Surogate key
Birth year	Artist.Birth_year	Dimension attribute
Death year	Atrist.Death_year	Dimension attribute
Nationality	Atrist.Nationality	Dimension attribute
Name	Atrist.Name&Surname	Dimension attribute

Dimension Definitions for Fact 2: Borrowed_or_lent

DIMENSION / ATTRIBUTE	TABLE.COLUMN	TYPE
Artwork	Artwork	Dimension
Title	Artwork.Title	Dimension attribute
Year created	Artwork.Year_created	Dimension attribute
Type	Artwork.Type	Dimension attribute
Medium	Artwork.Medium	Dimension attribute

Dimensions	Artwork.Dimensions	Dimension attribute
Permanent	Artwor ermane	Dimension attribute
Artwork l	Borrowed_or_lent.ID_Artwork	Surrogate key
Institution	Institutio	Dimension
Name	Institution.Name	Dimension attribute
Location	Institution.Location	Dimension attribute

Prestige	Institution. e	Dimension attribute
Institution ID	Borrowed_or_lent.ID _Institution	**Surrogate key**
Borrowing Status	Junk (Status)	Junk Dimension
Status	Junk.Status	Dimension attribute
Junk2 ID	Borrowed_or_lent.ID _Junk2	Surrogate key
Borrow Date	Date	Dimension
Year	Date.Year	Dimension attribute

Month	Date.Month	Dimension attribute
Working Day	Date.WorkingDay	Dimension attribute
Borrow Date ID	Borrowed_or_lent.ID_StartDate	Surrogate key
Lent Date	Date	Dimension
Year	Date.Year	Dimension attribute
Month	Date.Month	Dimension attribute
Working Day	Date.WorkingDay	Dimension attribute
Lent Date ID	Borrowed_or_lent.ID_StartDate	Surrogate key
Artist	Artist	Dimension
ID_Artist	Artist.ID_Artist	Surogate key
Birth year	Artist.Birth_year	Dimension attribute

Death year	Atrist.Death_ year	Dimension attribute
Nationality	Atrist.Nationality	Dimension attribute
Name	Atrist.Name&Surname	Dimension attribute

Checking the feasibility of queries based on the multidimensional model

1. Why was there an increase / decrease in donations this month?
Measure: Total amount of donations – SUM(Amount)

Dimension: Date (dimension attributes: Month, Year)

Dimension: Donation Purpose (from Junk)

Dimension: Artwork (dimension attributes: Title, Type)

2. Compare the total donations amount this month to the previous month.
Measure: Total amount of donations – SUM(Amount)

Dimension: Date (dimension attributes: Month, Year)

3. Have we got new patrons this month?
Measure: Number of new patrons – COUNT(ID_Patron)

Dimension: Patron (dimension attribute: Date_joined)

Dimension: Date (dimension attributes: Month, Year)

4. Who is the biggest patron this month?
Measure: Total amount of donations – SUM(Amount)

Dimension: Patron (dimension attributes: Name, Surname)

Dimension: Date (dimension attributes: Month, Year)

5. Which borrowed artwork earned the most?
Measure: Total amount of donations – SUM(Amount)

Dimension: Artwork (dimension attributes: Title)

Dimension: Borrowed_or_Lent (dimension attributes: Borrowed = TRUE)

Join Condition: Artwork must be marked as borrowed in Borrowed_or_Lent during the donation period

6. How much money came from donations for borrowed artworks?
Measure: Total amount of donations – SUM(Amount)

Dimension: Artwork (with FK to Borrowed_or_Lent)

Dimension: Borrowed_or_Lent (attribute: Borrowed = TRUE)

7. How Does Borrowing/Lending Art Influence Donations?
Measure: Total amount of donations – SUM(Amount)

Dimension: Borrowed_or_Lent (attributes: Borrowed, Lent, Owned)

Dimension: Date (dimension attributes: Year, Month)

8. How do donation amounts change when the museum exhibits borrowed artwork compared to its permanent collection?
Measure: Total amount of donations – SUM(Amount)

Dimension: Artwork (attribute: Permanent)

Dimension: Borrowed_or_Lent (attribute: Borrowed = TRUE)

Dimension: Date (dimension attributes: Month, Year)

9. Do donors contribute more when an artwork is on temporary loan rather than part of the museum's own collection?
Measure: Total amount of donations – SUM(Amount)

Dimension: Artwork (attribute: Permanent = FALSE)

Dimension: Borrowed_or_Lent (attribute: Borrowed = TRUE)

Dimension: Date (dimension attributes: Month, Year)

10. How does the prestige of the lending institution affect the frequency of donations?
Measure: Number of donation facts – COUNT(1)

Dimension: Institution (attribute: Prestige)

11. Does lending out the museum's own artwork to other institutions attract new patrons this year in comparison to last year?
Measure: Number of new patrons – COUNT(ID_Patron)

Dimension: Date (attribute: Year)

Dimension: Borrowed_or_Lent (attribute: Lent = TRUE)

12. How does the prestige of the institution that we lend out affect the total amount of donations?
Measure: Total amount of donations – SUM(Amount)

Dimension: Institution (attribute: Prestige)

Dimension: Borrowed_or_Lent (attribute: Lent = TRUE)

TABLE NAME: DONATION

COLUMN	SOURCE
INVOICE_NUMBER	Unique donation identifier. Taken from donations.csv.
ID_Date	ID_Date surrogate key - generated by the database, when donation is made.
ID_Patron	Foreign key to Patron. Taken from ID_Patron in donations.csv.

ID_Artwork	Foreign key to Artwork. Taken from ID_Artwork in artwork.csv.
ID_Junk1	Foreign key to Junk, taken from donations.csv.
Amount	Taken from donations.csv.
Tax	Taken from tax.csv.
Profit	Calculated as amount after subtracting tax.

TABLE NAME: PATRON

COLUMN	SOURCE
ID_Patron	ID_Patron surrogate key - generated by the database
Type	From patron.csv.
Phone_number	From patron.csv.

Address	From patron.csv.
Membership_status	From patron.csv.
Membership_level	From patron.csv.
Date_joined	From patron.csv.
Name	From patron.csv.
Surname	From patron.csv.

TABLE NAME: ARTWORK

COLUMN	**SOURCE**
ID_Artwork	ID_Artwork surrogate key - generated by the database
Title	From artwork.csv.

Year_created	From artwork.csv.
Type	From artwork.csv.
Medium	From artwork.csv.
Dimensions	From artwork.csv.
Permanent	From artwork.csv.

TABLE NAME: ARTIST

COLUMN	SOURCE
ID_Artist	ID_Artwork surrogate key - generated by the database
Birth_year	From artist.csv.

Death_year	From artist.csv.
Nationality	From artist.csv.
Name	From artist.csv.
Surname	From artist.csv.

TABLE NAME: INSTITUTION

COLUMN	SOURCE
ID_Institution	ID_Institution surrogate key - generated by the database
Name	From Borrowed_or_lend.csv.
Location	From Borrowed_or_lend.csv.
Prestige	From Borrowed_or_lend.csv.

TABLE NAME: BORROWED_OR_LENT

COLUMN	SOURCE
Borrowing_ID	Borrowing_ID surrogate key - generated by the database
ID_Artwork	Foreign key to Artwork. Taken from Artwork_ID.
Borrowed	From Borrowed_or_lend.csv.
Lent	From Borrowed_or_lend.csv.
Owned	From Borrowed_or_lend.csv.
ID_Junk2	From Borrowed_or_lend.csv.
ID_Institution	Foreign key to Institution. From Borrowed_or_lend.csv.

TABLE NAME: DATE

COLUMN	SOURCE
ID_Date	ID_Date surrogate key - generated by the database
Date	Standard date format (e.g., 2025-04-01). Generated by the database.
Year, Month, MonthNo	Extracted from Date.
DayOfWeek, DayOfWeekNo	Extracted from Date.
WorkingDay	Extracted from Date..
Vacation	Extracted from Date.
Holiday	Extracted from Date.